

Peripheral Ameloblastoma: A Case Report and Review of Literature.

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Peripheral ameloblastoma (PA) is a rare benign tumor that can occur mostly in the mandibular gingiva of the premolar area, originating from the residual odontogenic epithelium. The patient is usually asymptomatic and the lesion can be an occasional finding during routine intraoral examination. Due to the lack of clinical and radiographic pathognomonic features, the diagnosis is based on histopathological analysis, associated with 3D computed tomography (CT) imaging. Here, we report the case of a middle-aged man showing an asymptomatic, sessile, normochromic papillomatous mass of the lingual alveolar mucosa, in correspondence of tooth 4.4, which was histologically diagnosed as peripheral ameloblastoma. After the complete excision of the lesion, there were no recurrence and no symptoms during the 3-year follow-up. The treatment of choice for PA is a conservative surgical excision, which usually results in a good prognosis, together with a long-term follow-up, necessary to intercept possible recurrence or, more rarely, malignant transformation.

MAID chemotherapy regimen as a treatment strategy for metastatic malignant ameloblastoma: A case report.

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Ameloblastoma is generally characterized as a benign tumor originating in odontogenic epithelium. However, few cases of metastatic malignant ameloblastoma have also been reported. Due to the low incidence of malignant ameloblastoma, there is no established treatment regimen. To explore effective treatment for malignant ameloblastoma, we reported this case study. This report described a case of a 28-year-old malignant ameloblastoma female patient with multiple metastasis (brain and lung). The patient presented ameloblastoma of the left mandible in 2012. Three years later, local recurrence and brain metastasis was observed during a follow-up examination. Five years later, malignant ameloblastoma was detected by imaging and immunohistochemistry in the bilateral multiple pulmonary nodules and mediastinal lymph nodes. The patient was initially treated with tumor resection. Three years later after local recurrence and brain metastasis, she was accepted the extensive mandibulectomy supplemented with brain stereotactic body radiotherapy (SBRT). When diagnosed with pulmonary metastasis, the patient received combined chemotherapy regimen of MAID (mesna, adriamycin, ifosfamide and dacarbazine) for 6 cycles. The efficacy evaluation was partial remission (PR) after the 6 cycles of MAID. The last patient follow-up was July 24th 2018, and no evidence of progression was observed. The progression-free survival (PFS) of the patient was more than 9 months. Surgical resection is the optimal treatment for locally recurrent ameloblastoma. SBRT may be an effective treatment for unresectable oligometastasis of malignant ameloblastoma. Finally, combined chemotherapy of MAID showed encouraging effects in the management of metastatic malignant ameloblastoma.

[Malignant ameloblastoma. Apropos of a case].

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Ameloblastoma is a locally invasive, histologically nonmalignant tumor that may on very rare occasions give rise to metastases. A patient with a mandibular ameloblastoma presenting typical histologic

appearances developed a pulmonary metastasis confirmed by histology as arising from the primary tumor. Two groups of these extremely rare malignant ameloblastomas can be distinguished: those without histologic signs of degeneration that give rise to metastases identical to the primary lesion, and those with degenerative signs provoking metastases with the histologic appearance of undifferentiated carcinoma. No relation appears to exist between size of primary tumor, histologic type of ameloblastoma, or its course and tendency to produce metastases.

Anesthetic Challenges in Ameloblastoma of the Mandible: A Case Report.

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Ameloblastoma is a rare, locally aggressive benign tumor primarily affecting the mandible, with an incidence of 0.92 cases per million person-years and a male predominance. A two-year-old male presented with a right mandibular mass. CT imaging and histopathology confirmed ameloblastoma. He underwent wide local excision, mandibulectomy, and pectoralis major myocutaneous flap reconstruction under general anesthesia. Preoperative assessment revealed potential airway challenges; intubation was achieved with the backward, upward, rightward pressure (BURP) maneuver, and tracheostomy was performed to secure postoperative airway patency. The surgery was uneventful, and the patient was successfully weaned off ventilatory support by postoperative day four. This case underscores the importance of careful planning and expertise in pediatric ameloblastoma management, highlighting the effectiveness of direct laryngoscopy with BURP maneuver and prophylactic tracheostomy.

Multidisciplinary Approach to Mandibular Ameloblastoma: A Case Report on Surgical and Prosthetic Management.

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BACKGROUND Ameloblastoma is a locally aggressive, benign, odontogenic tumor. Reports suggest that the chances of recurrence of this tumor are high if treated with a conservative approach. Concordantly, surgical removal of the lesion along with the affected adjacent tissues and bone structure is recommended to reduce the chances of recurrence. Post-surgical prosthetic rehabilitation is advised to improve speech, mastication, and aesthetic appearance. This case report highlights the treatment and reconstruction challenges that maxillofacial surgeons and their teams face in managing cases of large ameloblastoma. **CASE REPORT** A 41-year-old Sudani man was referred for the management of a large ameloblastoma associated with the left border of the mandible. Management consisted of surgical removal of the affected mandible along with prosthetically preserving the mandible with grafts and screws. Histopathological, computed tomography, and incisional biopsy evaluation confirmed the presence of ameloblastoma. Postoperatively, no complications were reported. Six months postoperatively, no sign of recurrence was seen. The patient was referred to a surgeon for placement of an endosseous implant. **CONCLUSIONS** When dealing with large ameloblastoma, an interdisciplinary dental team is essential for improving the treatment results. This case highlights the importance of precise and timely primary care diagnosis and a collaborative approach to treatment. By embracing advancements in digital technologies, surgeons can enhance functional and aesthetic results, improving long-term quality of life.

Factors related to risk of recurrence and recurrence free survival in ameloblastoma of the Jaws: A single centre retrospective analysis.

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Ameloblastoma is a benign tumor originating from odontogenic epithelium with a global incidence of about 0.5 cases per million persons per year. The overall recurrence rate of ameloblastomas range from 55% -90%. In this retrospective study we have analyzed the factors associated with recurrence and recurrence free survival in ameloblastoma of the jaws. Retrospective chart review of 10 years was done from records directory. All demographic data, data on diagnostic modalities, clinical presentations, radiology, management and histology were taken into consideration. Kaplan Meir estimator was used to evaluate recurrence rate and Cox regression analysis (univariate and multivariate) was used to evaluate the odds ratio to find out the possible factors influencing risk of recurrence and influence recurrence free survival. ROC curve (Receiver Operating Characteristic) was used to find out the optimal cut off point for size of the lesion in predicting recurrence. A p value of < 0.05 was considered statistically significant at 95% confidence interval. A total of 75 patients had recurrence with a median follow up of 65 months. The overall 5 year RFS was 19.5%. Conventional ameloblastomas, cortical/ soft tissue invasion, conservative treatment and tumor size ≥ 4 cm were independent predictors of recurrence. Aggressive treatment is preferred for conventional ameloblastomas ≥ 4 cm with cortical/ soft tissue invasion.

Metabolic Analysis of Tumor Cells Within Ameloblastoma at the Single-Cell Level.

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To meet their high energy needs, tumor cells undergo aberrant metabolic reprogramming. A tumor cell may expertly modify its metabolic pathways and the differential expression of the genes for metabolic enzymes. The physiological requirements of the host tissue and the tumor cell of origin mostly dictate metabolic adaptation. Ameloblastoma (AB) is a benign odontogenic tumor of epithelial origin. Due to its unrestricted growth potential, local aggressiveness, and high likelihood of recurrence, this condition poses a significant risk to the patient's health. This study aimed to characterize the metabolic heterogeneity at single-cell resolution of AB. Single-cell RNA sequencing (scRNA-seq) was performed on 17,284 cells from three AB donors. Bioinformatic analysis was used to examine differentially expressed genes, subtypes, and regulatory mechanisms when combined with odontogenic keratocyst scRNA-seq data. Based on metabolic pathway gene sets, the metabolic landscape of AB tumor cells was examined. Using scRNA-seq, we discovered that AB tumor cells had substantial heterogeneity. The biggest contributor to tumor cell metabolic characteristics is determined to be variation in mitochondrial programming and glycolysis. Surprisingly, hypoxia corresponds with both oxidative phosphorylation and glycolysis activity in AB tumor cells at the single-cell level. This study presents a computational framework for defining metabolism using single-cell expression data and identifies oxidative phosphorylation and glycolysis as critical components of metabolism for AB tumor cells.

MMP13 Expression and Activity Suggest Its Role in Bone Resorption in Ameloblastomas.

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Ameloblastoma is a locally destructive benign odontogenic tumor. While the neoplastic cells of

conventional ameloblastoma can infiltrate the connective tissue and bone, in unicystic ameloblastoma the epithelium is encapsulated. The mechanisms driving ameloblastoma's bone resorption remains unclear. RNA sequencing (RNA-seq) was performed in a discovery cohort of conventional ameloblastoma, and pathway enrichment analysis was carried out. mRNA levels of MMP13, a gene associated with bone resorption, were assessed using RT-qPCR in a larger cohort of conventional ameloblastoma and in unicystic ameloblastoma. Zymogram gels and the immunoexpression profile of collagenase 3 (encoded by MMP13 gene) were evaluated as well. Enriched pathways related to bone mineralization and upregulation of MMP13 were observed in ameloblastomas. Collagenolytic activity of collagenase 3 was detected in the tumor lysates. Collagenase 3 immunopositivity was observed in ameloblastomatous epithelium infiltrating the fibrous capsule of unicystic ameloblastoma. At the tumor-bone interface, collagenase 3 expression was detected in stromal cells, osteoblasts, and osteocytes. The results indicate a potential involvement of MMP13 in ameloblastoma-related bone resorption and progression.

Benign metastasizing ameloblastoma or malignant ameloblastoma?

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Ameloblastoma of the maxilla is a rare odontogenic tumor that rarely metastasizes. We report a patient who was diagnosed with ameloblastoma of the maxilla 6 years ago and had undertaken the operation. However, the recurrence occurred, and further pulmonary disease was discovered. He did not experience chest pain and other respiratory systematic symptoms. Whether this metastasizing ameloblastoma should be ranged among malignant or benign is discussed. The term benign metastasizing ameloblastoma is suggested to describe this kind of tumor.

Cytology of recurrent ameloblastoma with malignant change. A case report.

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Ameloblastoma is a rare tumor of the jaw that arises from the odontogenic epithelium. Ameloblastomas have a propensity for local recurrence and, rarely, for metastasis. The term malignant ameloblastoma is reserved for those metastasizing tumors that retain the typical morphology of ameloblastoma. Fine needle aspiration (FNA) reports on ameloblastomas are scant, and those on malignant ameloblastomas are still more so. In a case of malignant ameloblastoma diagnosed by FNA cytology, the clinical presentation was that of a malignant neoplasm. FNA smears were highly cellular and showed isolated, scattered cells and small groups of basaloid cells and polygonal squamous epithelial cells. Stellate and spindle-shaped cells were also seen in the background. The cytologic diagnosis was subsequently confirmed on histopathology. The characteristic combination of cells in FNA smears facilitated the diagnosis of ameloblastoma. Since the biologic behavior of the tumor was that of a malignant neoplasm, the slides were reviewed. The cytologic smears did not exhibit sufficient features of malignancy to label the lesion malignant.